A Literature Survey of Hybrid Big Data and Quantum Computing for Smart Delivery Route Optimization

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Abstract

The increasing volume of delivery, traffic, and road-network data has created new challenges for efficient route optimization. Traditional routing approaches such as the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP) become computationally difficult as the number of vehicles, customers, and routing constraints increases. Recent research has therefore investigated Big Data processing, traffic-aware routing, and quantum computing as possible approaches to address these challenges. This literature survey reviews research on Big Data stream processing, dynamic traffic-aware route selection, quantum optimization for vehicle routing, and systematic studies of quantum computing for routing problems. Particular attention is given to QUBO formulations, quantum annealing, the Quantum Approximate Optimization Algorithm (QAOA), and hybrid quantum-classical approaches. The reviewed studies indicate that quantum computing shows potential for combinatorial routing problems, while current hardware limitations, problem size, noise, and parameter selection remain important challenges. Based on the reviewed literature, a hybrid architecture combining Big Data processing, traffic-aware routing, classical optimization, and QAOA is identified as a promising direction for smart delivery route optimization.

***Index Terms*—**Big Data, vehicle routing problem, quantum computing, QAOA, QUBO, quantum annealing, traffic-aware routing, delivery route optimization, hybrid quantum-classical computing.

I INTRODUCTION

Routing optimization is an important problem in transportation and logistics because an efficient route can reduce travel distance, travel time, fuel consumption, and delivery delays. However, routing problems become computationally difficult as the number of customers, vehicles, and constraints increases. The Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP) are among the most widely studied routing problems because of their computational complexity and practical applications in logistics and transportation [1].

Recent research has explored quantum computing as an alternative approach for solving complex routing and optimization problems. A systematic literature review of quantum computing for routing problems analyzed 53 studies published over an 18-year period. The review found that TSP was the most frequently studied routing problem, followed by VRP. It also showed increasing interest in hybrid quantum-classical approaches as quantum hardware developed [1].

At the same time, transportation systems generate large volumes of data from traffic sensors, vehicles, road networks, and delivery operations. Big Data technologies can therefore be used to process and analyze transportation information before performing route optimization. Research on real-time Big Data analytics has investigated distributed processing and streaming architectures for handling continuously arriving data [2].

Traffic conditions are another important factor in delivery routing. A route that is shortest under normal conditions may not be the fastest when congestion is present. Research on dynamic route selection has therefore considered travel time, distance, road conditions, congestion, and other factors when selecting routes [3].

Quantum optimization approaches provide another direction for routing. In these approaches, a routing problem can be converted into a mathematical representation such as a Quadratic Unconstrained Binary Optimization (QUBO) model. The resulting optimization problem can then be addressed using quantum annealing or gate-based methods such as the Quantum Approximate Optimization Algorithm (QAOA) [4].

The reviewed literature also shows that current quantum computers have limitations related to noise, decoherence, available qubits, connectivity, and computational resources. Consequently, hybrid quantum-classical approaches are particularly important because classical computers can perform parts of the optimization while quantum methods are used for selected optimization tasks [5].

This literature survey therefore focuses on four major areas: Big Data processing for transportation, traffic-aware route

optimization, quantum computing for routing, and hybrid quantum-classical optimization. The objective is to identify the contributions and limitations of existing research and to determine a suitable research direction for smart delivery route optimization.

II LITERATURE REVIEW METHODOLOGY

The literature considered in this study focuses on the application of Big Data processing, intelligent transportation systems, and quantum computing to route optimization and vehicle routing problems. The selected studies were examined to understand the problem addressed, optimization technique, data processing approach, quantum computing method, experimental environment, and limitations.

The reviewed literature can be broadly divided into four categories. The first category focuses on Big Data processing and real-time analytics for transportation systems. These studies emphasize the need to process large volumes of continuously generated data using distributed and streaming technologies [2]. The second category focuses on traffic-aware route selection, where changing traffic conditions are incorporated into the route optimization process [3].

The third category investigates quantum computing for vehicle routing problems. These studies formulate routing problems using binary optimization models and investigate quantum optimization methods such as quantum annealing and the Quantum Approximate Optimization Algorithm (QAOA) [4]. The fourth category consists of survey and systematic review studies that analyze the overall development of quantum computing in transportation and routing problems [1], [5].

II-A Selection of Literature

The literature was selected based on its relevance to the objectives of smart delivery route optimization. Particular attention was given to studies addressing vehicle routing, traffic-aware routing, Big Data processing, QUBO formulation, quantum optimization, and hybrid quantum-classical approaches.

The selected papers include research studies proposing routing algorithms as well as survey papers describing the development of quantum computing in transportation. This combination is important because research papers provide specific algorithms and experimental results, while survey papers provide a broader understanding of the research field and its limitations.

II-B Analysis Criteria

Each selected study was analyzed according to the following criteria:

- problem addressed;
- routing model or transportation application;
- dataset or experimental environment;
- classical or quantum algorithm used;
- software libraries and technologies;
- optimization formulation;
- performance measures;
- major findings; and
- limitations and identified research gaps.

These criteria allow the reviewed studies to be compared in a consistent manner and help identify the techniques that are most relevant to the proposed smart delivery route optimization system.

II-C Research Problems Addressed

Routing problems are computationally challenging because the number of possible routes increases rapidly as the number of customers and vehicles increases. The systematic literature review by Osaba *et al.* analyzed 53 studies and found that the Traveling Salesman Problem (TSP) was the most frequently studied routing problem, while the Vehicle Routing Problem (VRP) was the second major category [1].

The review reported that 32 of the 53 studies focused on TSP, while 13 studies addressed VRP or its variants. Other studies considered shortest-path, Hamiltonian-cycle, and graph traversal problems [1]. These findings indicate that routing optimization is an important application area for quantum computing.

For delivery applications, the VRP is particularly relevant because it considers multiple vehicles serving multiple customers. Additional constraints such as vehicle capacity and delivery time restrictions can further increase the complexity of the problem. Existing quantum-routing studies have therefore investigated both standard and constrained versions of routing problems.

II-D Classical and Quantum Approaches

Traditional routing methods rely mainly on classical optimization, heuristic, and metaheuristic techniques. However, the reviewed literature shows increasing interest in quantum approaches for combinatorial optimization. Two major quantum-computing architectures appear in the routing literature: quantum annealing and gate-model quantum computing.

Quantum annealing attempts to obtain a low-energy state corresponding to a good solution of an optimization problem. Gate-model approaches instead construct quantum circuits using quantum gates to manipulate qubit states [1].

Recent studies increasingly use hybrid quantum-classical approaches. In such approaches, the classical computer performs part of the optimization while the quantum processor is used for the quantum optimization component. This approach is particularly important for current quantum hardware because available quantum devices have limitations in terms of qubit count, noise, and computational reliability.

The selected literature represents the major components required for the development of a smart delivery route optimization system. The studies were analyzed according to the transportation problem addressed, optimization algorithm, data-processing approach, quantum technique, experimental environment, and reported limitations.

The first research direction concerns the processing of large amounts of transportation data. The PAQS study investigates real-time Big Data analytics using distributed and streaming

TABLE I
RESEARCH PAPERS CONSIDERED IN THIS LITERATURE SURVEY

Paper	Main Problem	Main Approach
PAQS	Real-time Big Data analytics	Kafka, Spark, quantum-inspired processing
IQGA	Dynamic vehicle route selection	Quantum Genetic Algorithm
QAOA-VRP	Vehicle Routing Problem	QUBO and QAOA
Transportation Survey	Quantum computing in transportation	QAOA, quantum annealing, QML, and other approaches
Routing Literature Review	Quantum computing for routing	Systematic analysis of TSP, VRP, and quantum approaches

processing techniques. Its approach demonstrates the importance of processing continuously generated data before performing intelligent analysis [2].

The second research direction focuses on traffic-aware route selection. The IQGA study considers the effect of changing traffic conditions on route selection and uses a quantum genetic algorithm to search for suitable routes. The study considers factors such as distance, travel time, congestion, road conditions, and other route characteristics when evaluating alternatives [3].

The third direction is the formulation of vehicle routing as a quantum optimization problem. The QAOA-based VRP study converts the routing problem into a Quadratic Unconstrained Binary Optimization (QUBO) formulation. The resulting problem is then converted into an Ising-type representation and solved using the Quantum Approximate Optimization Algorithm. A classical optimizer is used to adjust the QAOA parameters, making the approach hybrid in nature [4].

The transportation survey provides a broader view of how quantum computing is being applied to transportation problems. It discusses quantum annealing, QAOA, quantum machine learning, quantum random walks, and other quantum approaches. It also highlights practical challenges such as limited qubit counts, noise, decoherence, connectivity, and the difficulty of scaling quantum algorithms to realistic transportation problems [5].

A systematic literature review of quantum computing for routing problems provides further evidence of the development of this research area. The review analyzed 53 studies and found that the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP) are among the most frequently investigated routing problems. It also identified both pure quantum and hybrid quantum-classical approaches in the existing literature [1].

III ANALYSIS OF THE REVIEWED LITERATURE

The reviewed studies can be organized into three closely related areas: Big Data processing, traffic-aware route optimization, and quantum-based routing optimization. These areas address different stages of the overall delivery-routing problem.

Big Data processing provides the data-processing foundation. Traffic-aware optimization uses this information to estimate changing route costs, while quantum optimization can be applied to the resulting combinatorial routing problem.

III-A Big Data Processing

Transportation systems continuously generate information from traffic sensors, vehicles, road networks, and other sources. Processing such data using conventional single-machine methods can become difficult when the volume and rate of incoming data increase. The PAQS study therefore investigates distributed and streaming approaches for real-time Big Data analytics [2].

For the proposed delivery-routing application, this concept is important because traffic information may change while delivery vehicles are operating. Instead of treating traffic as a fixed input, a streaming architecture can continuously update the information used by the routing system.

III-B Traffic-Aware Routing

A shortest-distance route is not necessarily the fastest route when traffic congestion changes travel conditions. The IQGA study addresses this issue by considering both relatively static route characteristics and dynamic traffic-related costs when selecting routes [3].

This provides an important basis for a smart delivery system. Distance, travel time, congestion, fuel consumption, and delivery delay can be combined into a route cost. The route optimization algorithm can then search for routes that minimize the overall cost rather than minimizing distance alone.

III-C Quantum Optimization for Routing

The routing problem becomes more difficult when multiple vehicles and customers are considered simultaneously. The QAOA-based VRP study addresses this problem by representing routing decisions using binary variables and incorporating the objective and constraints into a QUBO formulation [4].

The QUBO formulation provides a bridge between the transportation problem and quantum optimization. After transformation into an Ising Hamiltonian, QAOA can be used to search for low-cost solutions. Because the parameters of QAOA are optimized using a classical optimizer, the overall procedure is a hybrid quantum-classical process.

III-D Evolution of Research in Quantum-Based Routing

The reviewed literature shows a gradual development from large-scale transportation data processing toward intelligent and quantum-assisted route optimization. The research can be viewed as a progression in which different studies address different components of the overall transportation optimization problem.

Early work in the selected literature focuses on the use of quantum-inspired techniques for transportation and traffic

management. The IQGA study investigates dynamic route selection in urban traffic and applies a quantum genetic algorithm to evaluate alternative routes under changing traffic conditions [3].

Subsequent research has investigated the direct application of quantum optimization to vehicle routing problems. The QAOA-based VRP study formulates the Vehicle Routing Problem as a QUBO model and applies the Quantum Approximate Optimization Algorithm through a hybrid quantum-classical optimization process [4].

More recent work has also considered the importance of processing large volumes of continuously generated data. The PAQS study investigates Big Data stream processing using technologies such as Kafka and Spark, providing a basis for real-time data processing before optimization [2].

Survey and systematic review studies provide a broader view of these developments. The transportation survey discusses quantum annealing, QAOA, quantum machine learning, quantum random walks, and other approaches for transportation applications [5]. Similarly, Osaba *et al.* systematically reviewed quantum computing for routing problems and analyzed the development of TSP, VRP, and other routing applications [1].

The evolution identified from the selected literature can therefore be summarized as follows.

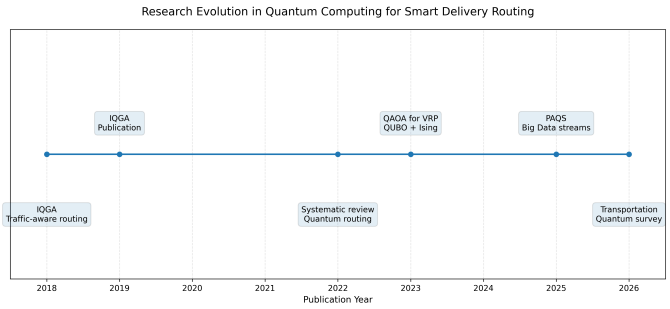


Fig. 1. Evolution of the reviewed research toward hybrid quantum-assisted delivery route optimization.

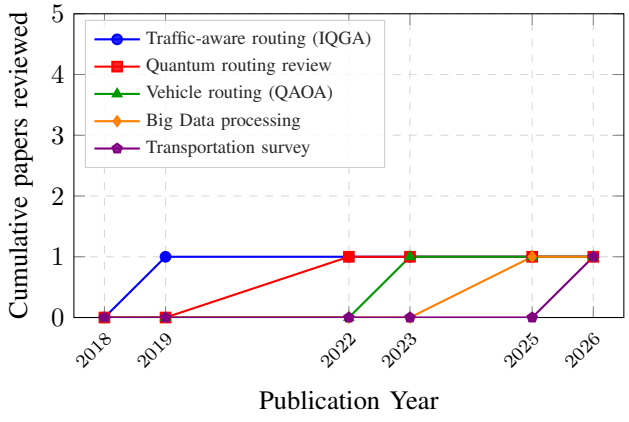


Fig. 2. Timeline of the reviewed papers by research category, 2018–2026.

The evolution demonstrates that the different studies address complementary aspects of the proposed research problem. Big Data research addresses the collection and processing of transportation information, traffic-aware routing considers changing road conditions, and quantum optimization addresses the computational complexity of routing problems.

III-E Types of Routing Problems Studied

Routing problems represent an important class of combinatorial optimization problems in transportation. The selected literature covers several routing formulations, with particular emphasis on the Traveling Salesman Problem (TSP) and Vehicle Routing Problem (VRP).

The TSP considers a single route that visits a set of locations and returns to the starting location. The objective is generally to determine an efficient tour through all required locations. The systematic literature review by Osaba *et al.* found that TSP was the most frequently investigated routing problem among the studies analyzed in their review [1].

The Vehicle Routing Problem extends the routing concept to multiple vehicles. In a typical VRP, a fleet of vehicles starts from a depot and must serve a set of customers while satisfying specified constraints. This formulation is more closely related to real-world delivery systems than the basic TSP because it explicitly considers multiple vehicles.

Several variants of VRP introduce additional constraints. For example, the Capacitated Vehicle Routing Problem (CVRP) considers the capacity of each vehicle, while other variants may include time windows, service requirements, or dynamic changes in travel conditions.

The QAOA-based routing study specifically investigates VRP and represents routing decisions using binary variables. The objective and routing constraints are incorporated into a QUBO formulation, which is subsequently used for quantum optimization [4].

The systematic review reported that VRP was the second major routing problem category after TSP. It also identified variants of VRP involving additional constraints and different practical applications [1].

For smart delivery applications, VRP is therefore more relevant than the basic TSP because a delivery system normally involves multiple vehicles and multiple customers. When traffic conditions are included, the problem becomes more dynamic because the cost of traveling along a road can change with time.

III-F Relationship Between Traffic and Routing

The reviewed studies indicate that route optimization should not consider distance alone. Traffic conditions can change travel time and consequently affect the quality of a selected route. The IQGA study incorporates traffic-related factors into route selection and evaluates alternative routes using preference and cost components [3].

This idea can be extended to delivery routing by defining a combined route cost based on factors such as distance, travel time, congestion, fuel consumption, and delivery delay.

Such a cost can then be incorporated into the optimization formulation.

Therefore, the literature suggests a connection between real-time transportation data and quantum optimization:

$$\text{Transportation Data} \rightarrow \text{Traffic-Aware Cost} \rightarrow \text{VRP} \rightarrow \text{QUBO} \rightarrow \text{QAOA} \quad (1)$$

This connection provides the basis for integrating Big Data processing with quantum-assisted delivery route optimization.

To make the relationship between the selected studies and their underlying techniques easier to follow, the five papers can also be organized according to the specific transportation problem category they address. This classification complements Table III by grouping each study under a single research category, along with the principal technique associated with that category. Such a grouping is useful because it shows, at a glance, which stage of the overall smart delivery pipeline each study contributes to, ranging from raw data processing through traffic-aware cost estimation to the final quantum optimization step.

Table IV presents this categorization. The Big Data category is represented solely by the PAQS study, which relies on Kafka and Spark for distributed stream processing. The Traffic Routing category corresponds to the IQGA study, which applies a Quantum Genetic Algorithm to incorporate dynamic traffic conditions into route selection. The Vehicle Routing category is represented by the QAOA-based VRP study, which relies on a QUBO and Ising formulation solved using QAOA. The two remaining categories, Transportation Survey and Routing Review, correspond to the broader survey and systematic review papers, which discuss a wider range of techniques including quantum annealing and quantum machine learning rather than a single algorithmic approach.

This classification highlights an important structural pattern in the reviewed literature: each individual research paper tends to focus on a single stage of the overall transportation pipeline, while the survey and review papers are the only sources that span multiple stages simultaneously. This pattern reinforces the observation made earlier in this survey that Big Data processing, traffic-aware routing, and quantum optimization have largely been investigated as separate research threads rather than as parts of a single integrated system.

The classification in Table IV also provides a convenient basis for identifying which techniques would need to be combined in order to build a complete smart delivery route optimization system. In particular, moving from the Big Data category to the Vehicle Routing category requires bridging streaming data processing with a QUBO-based problem formulation, while moving from the Traffic Routing category to the Vehicle Routing category requires incorporating dynamic traffic-related costs directly into the QUBO objective function. The survey and review papers, although not tied to a specific implementation, provide the broader technical context needed to make these connections, since they summarize the range of quantum algorithms, hardware platforms, and evaluation practices that have been used across the field.

The comparison presented in Table III shows that the selected studies address different but complementary aspects of transportation route optimization. The PAQS study concentrates on the processing of large-scale and continuously generated data, while the IQGA study focuses on selecting routes under changing urban traffic conditions [2], [3].

The QAOA-based VRP study moves from traffic and data processing toward the direct optimization of vehicle routes. It represents routing decisions using binary variables and formulates the objective and constraints as a QUBO problem. QAOA is then applied using a hybrid quantum-classical procedure [4].

The survey studies provide a broader perspective on the research field. The transportation survey discusses several quantum approaches, including QAOA, quantum annealing, and quantum machine learning, while also identifying limitations associated with current quantum hardware and scalability [5].

The systematic literature review by Osaba et al. further shows that TSP and VRP are major research areas within quantum routing. The review analyzed 53 studies and identified both quantum and hybrid quantum-classical approaches [1].

An important observation from the reviewed literature is that the three major components of the proposed research are generally investigated separately. Big Data approaches address the processing of transportation information, traffic-aware methods address changing road conditions, and quantum optimization approaches address computationally difficult routing problems.

This observation suggests the need for an integrated framework that can combine real-time transportation data with traffic-aware multi-vehicle routing and quantum optimization. Such integration forms the basis for the proposed smart delivery route optimization approach.

III-G Algorithms and Quantum Computing Approaches

The reviewed literature contains both quantum-inspired and gate-based quantum optimization approaches. The IQGA study uses a quantum genetic algorithm for dynamic route selection. Although the method uses quantum concepts, its experimental quantum process is simulated using classical computing resources [3].

Quantum annealing represents another important approach in the transportation literature. It converts optimization problems into suitable energy-based formulations and searches for low-energy solutions. Quantum annealing has been investigated for routing, scheduling, and traffic-control problems.

Gate-based quantum computing provides another approach through algorithms such as QAOA. QAOA is particularly relevant to the proposed research because it can operate on a QUBO representation of the vehicle routing problem. The QAOA-based VRP study demonstrates this process using a hybrid quantum-classical optimization procedure [4].

TABLE II
DETAILS OF THE RESEARCH AND SURVEY PAPERS CONSIDERED

ID	Reference	Year	Problem	Approach
1	PAQS: Advanced Optimization for Big Data Streams with Quantum Insights for Real-time Big Data Analytics	2025	Real-time Big Data analytics	Kafka, Spark, distributed processing and quantum-inspired optimization
2	IQGA: A Route Selection Method Based on Quantum Genetic Algorithm Toward Urban Traffic Management Under Big Data Environment	2018	Dynamic vehicle route selection	Quantum Genetic Algorithm with traffic-aware route evaluation
3	Solving Vehicle Routing Problem Using Quantum Approximate Optimization Algorithm	2023	Vehicle Routing Problem	QUBO formulation, Ising model and QAOA
4	Quantum Computing in Transportation Engineering: A Survey	–	Transportation optimization	Review of QAOA, quantum annealing, QML and other approaches
5	A Systematic Literature Review of Quantum Computing for Routing Problems	2022	TSP, VRP and routing problems	Systematic review of quantum and hybrid quantum-classical approaches

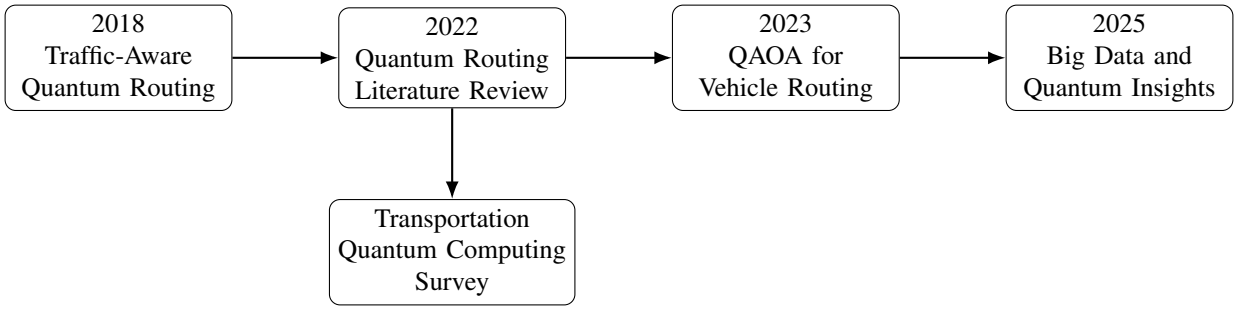


Fig. 3. Evolution of the selected research toward hybrid quantum-assisted transportation and delivery route optimization.

The systematic literature review also shows that different quantum platforms have been used in routing research, including D-Wave quantum annealers and IBM gate-model quantum computers. However, the use of quantum hardware does not by itself establish quantum advantage. Performance must be evaluated against strong classical approaches under comparable conditions [1].

The choice between quantum annealing and gate-model computing depends on the way the routing problem is formulated. Quantum annealing is generally suited to problems that can be expressed directly as a QUBO or Ising energy function, since the annealer searches for a low-energy configuration corresponding to a good routing solution. Gate-model approaches such as QAOA instead require the problem to be encoded into a parameterized quantum circuit, with a classical optimizer iteratively adjusting the circuit parameters to minimize the expected cost. Both approaches ultimately rely on the same underlying QUBO representation of the vehicle routing problem, but differ in how the optimization is executed on the underlying hardware.

Software toolkits also play an important role in the reviewed literature. Gate-based studies commonly use frameworks such as Qiskit to construct and simulate quantum circuits before executing them on physical hardware, while annealing-based studies rely on vendor-specific software provided by D-Wave. The availability of such toolkits has made it easier for transportation researchers without a specialized quantum computing background to experiment with QUBO formulations of routing problems, which partly explains the growing number of studies applying quantum optimization to TSP and VRP variants.

An additional consideration raised in the reviewed literature is the trade-off between solution quality and computational overhead. Although quantum annealers can evaluate many candidate solutions in parallel through quantum tunneling, the mapping of a routing problem onto the physical qubit topology of the annealer can introduce additional overhead through minor embedding, which may reduce the effective size of the problem that can be solved. Gate-model approaches face a related limitation because deeper circuits are more expressive but are also more susceptible to noise on current hardware. These factors reinforce the conclusion that hybrid quantum-classical approaches, which limit the size and depth of the quantum component, remain the most practical direction for

TABLE III
COMPARISON OF THE SELECTED RESEARCH AND SURVEY PAPERS

ID	Research Work	Problem	Algorithm / Method	Main Contribution
1	PAQS (2025)	Real-time Big Data analytics	Kafka, Spark, distributed and quantum-inspired processing	Provides an approach for processing large-scale and streaming data for real-time analytics.
2	IQGA (2018)	Dynamic urban route selection	Quantum Genetic Algorithm	Considers traffic-related factors when selecting routes in urban transportation networks.
3	QAOA for VRP (2023)	Vehicle Routing Problem	QUBO, Ising model, QAOA	Formulates VRP as a QUBO problem and applies a hybrid quantum-classical optimization process.
4	Quantum Computing in Transportation Engineering	Transportation problems	QAOA, quantum annealing, QML and other approaches	Surveys applications and identifies hardware, software, and scalability challenges.
5	Osaba et al. (2022)	TSP, VRP and routing problems	Systematic literature review	Analyzes the development of quantum computing research for routing problems.

TABLE IV
CLASSIFICATION OF THE SELECTED STUDIES

Category	Studies	Main Technique
Big Data	PAQS	Kafka, Spark, distributed processing
Traffic Routing	IQGA	Quantum Genetic Algorithm
Vehicle Routing	QAOA-VRP	QUBO, Ising, QAOA
Transportation Survey	Transportation Survey	QAOA, quantum annealing, QML
Routing Review	Osaba et al.	TSP, VRP, quantum and hybrid methods

routing problems of realistic size.

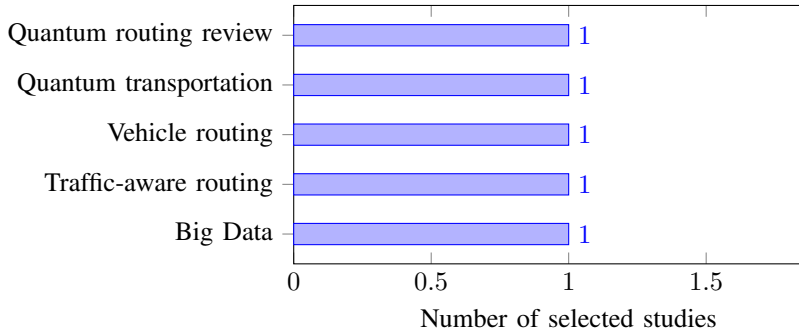


Fig. 4. Research focus of the selected studies reviewed in this work.

III-H Research Focus of the Reviewed Studies

Figure 4 summarizes the main research focus of the selected studies. The reviewed literature covers different stages of transportation route optimization. The Big Data study focuses on processing large-scale and continuously generated data, while the traffic-aware routing study considers changing traffic conditions during route selection.

The vehicle routing study focuses on the mathematical and quantum optimization of routing decisions. In particular, the routing problem is formulated as a QUBO model and solved using QAOA through a hybrid quantum-classical procedure [4].

The two review studies provide a broader perspective of quantum computing in transportation and routing. They show that several quantum approaches have been investigated, including quantum annealing, QAOA, quantum-inspired algorithms, and quantum machine learning [1], [5].

The combination of these studies indicates that transportation optimization can be viewed as a sequence of interconnected problems. Data must first be collected and processed, traffic conditions must be incorporated into the route cost, and the resulting routing problem must then be optimized.

III-I Research Impact and Development

The reviewed literature demonstrates increasing interest in applying quantum computing to transportation and routing problems. The systematic literature review by Osaba *et al.* analyzed 53 studies published between 2004 and 2021. Among these studies, TSP was the most frequently investigated routing problem, followed by VRP [1].

The review also identified the use of both pure quantum and hybrid quantum-classical approaches. This indicates that research has not been limited to theoretical quantum algorithms, but has also investigated combinations of classical optimization and quantum computing [1].

For practical transportation applications, hybrid approaches are particularly relevant because current quantum hardware has limitations in terms of the number of available qubits, noise, connectivity, and circuit depth. The transportation survey emphasizes these limitations and identifies scalability as one of the major challenges in applying quantum computing to realistic transportation systems [5].

Therefore, the current literature suggests that the most practical research direction is not to completely replace classical transportation algorithms with quantum computers. Instead, quantum computing can be integrated with classical methods to solve selected computationally difficult components of a larger transportation system.

III-J Open Challenges in Quantum-Based Transportation

Although the reviewed studies demonstrate the potential of quantum computing for transportation optimization, several challenges remain.

- **Scalability:** Many quantum routing experiments are performed on relatively small problem instances. Scaling the problem to realistic numbers of vehicles and customers remains difficult.
- **Quantum hardware limitations:** Current quantum devices are affected by noise, decoherence, limited qubit counts, and restricted connectivity.
- **Dynamic transportation conditions:** Traffic conditions can change continuously. A routing solution that is optimal at one time may become inefficient when congestion changes.
- **Integration with Big Data:** Large-scale transportation data must be processed efficiently before it can be incorporated into an optimization model.
- **Constraint handling:** Real delivery systems involve vehicle capacities, customer requirements, delivery deadlines, and depot constraints. Encoding these constraints into quantum optimization models can increase the complexity of the problem.
- **Classical comparison:** Quantum approaches need to be evaluated against strong classical optimization methods using comparable datasets, constraints, and computational budgets.

III-K Major Approaches Identified

The selected literature contains several distinct approaches to transportation optimization. The PAQS study represents the Big Data processing perspective, emphasizing streaming and distributed processing for large-scale analytics [2].

The IQGA study represents the traffic-aware optimization perspective. It considers dynamic traffic conditions and applies a quantum genetic algorithm for route selection [3].

The QAOA-based VRP study represents the gate-based quantum optimization perspective. It demonstrates how a vehicle routing problem can be transformed into QUBO and subsequently addressed using QAOA [4].

The survey and systematic review studies provide evidence that these approaches form part of a broader research area involving quantum annealing, QAOA, quantum-inspired optimization, and quantum machine learning [1], [5].

III-L Research Gap

The reviewed literature reveals that important components of smart transportation optimization have been investigated, but they are often addressed independently. Big Data research primarily focuses on data processing, traffic-aware research focuses on dynamic route selection, and quantum optimization research focuses on solving combinatorial routing problems.

A further limitation is that many quantum routing studies evaluate relatively small or simplified problem instances. The systematic review emphasizes the need for more realistic routing applications and stronger evaluation of quantum approaches against classical optimization techniques [1].

Similarly, the transportation survey identifies scalability, hardware limitations, noise, and the need for standardized benchmarking as important open challenges [5].

Therefore, there is a research opportunity to investigate an integrated framework that combines large-scale transportation data processing, traffic-aware cost estimation, multi-vehicle delivery routing, and hybrid quantum-classical optimization.

Based on these observations, the proposed research focuses on the following pipeline:

$$\text{Big Data} \rightarrow \text{Traffic-Aware Cost} \rightarrow \text{VRP} \rightarrow \text{QUBO} \rightarrow \text{QAOA}. \tag{2}$$

The objective is to investigate whether a hybrid approach can produce competitive solutions for smart delivery route optimization while maintaining comparison with established classical optimization methods.

III-M Hybrid Architecture for Delivery Route Optimization

The reviewed literature indicates that hybrid quantum-classical approaches are particularly important for current transportation applications. Classical computers can perform data processing, problem formulation, and parameter optimization, while the quantum processor is used for a selected optimization component. This reduces the need to perform the complete transportation workflow on current quantum hardware. Figure 5 summarizes how classical, quantum, and

TABLE V
ALGORITHMS, COMPUTING PLATFORMS, AND PROBLEMS IN THE REVIEWED STUDIES

ID	Study	Algorithm / Method	Platform / Technology	Problem
1	PAQS	Quantum-inspired optimization	Kafka, Spark, distributed processing	Real-time Big Data analytics
2	IQGA	Quantum Genetic Algorithm	Classical simulation	Dynamic urban route selection
3	QAOA-VRP	QUBO, Ising, QAOA	Qiskit / quantum simulation	Vehicle Routing Problem
4	Transportation Survey	QAOA, quantum annealing, QML, quantum random walk	IBM, D-Wave and other quantum platforms	Transportation optimization
5	Osaba et al.	Quantum annealing, gate-model and hybrid methods	D-Wave, IBM and others	TSP, VRP and routing problems

TABLE VI
MAJOR LIMITATIONS IDENTIFIED IN THE REVIEWED LITERATURE

Challenge	Description
Scalability	Many quantum routing experiments use small problem instances, making large-scale delivery optimization difficult.
Noise	Current quantum devices are affected by computational errors and noise.
Limited qubits	The number of available qubits restricts the size of problems that can be directly represented.
Parameter selection	QAOA performance depends on suitable circuit parameters and classical optimization.
Dynamic traffic	Changing traffic conditions require route costs to be updated continuously.
Constraint handling	Vehicle capacity, delivery deadlines, and depot constraints increase the complexity of quantum formulations.
Classical comparison	Quantum methods need comparison with strong classical optimization methods.

hybrid approaches relate to one another for the proposed delivery-routing problem.

III-N Challenges Identified From the Literature

Although quantum computing provides promising techniques for routing optimization, the reviewed studies reveal several limitations. The first major challenge is scalability. Many experiments are conducted on small routing instances because current quantum devices provide a limited number of usable qubits.

Noise and decoherence are additional challenges for gate-based quantum computing. Errors introduced during quantum operations can affect the measured solution and make reliable optimization difficult. The transportation survey therefore identifies error mitigation and improved quantum hardware as important directions for future research [5].

Another challenge is parameter optimization. QAOA requires suitable values for its parameters, and the quality of the result can depend strongly on the selected circuit depth and optimization procedure. Increasing the QAOA depth may improve the expressive capability of the algorithm but also increases circuit complexity and exposure to noise.

Real-world delivery routing introduces further challenges. Unlike small benchmark problems, practical delivery systems may involve many vehicles, customer demands, vehicle capacities, delivery deadlines, and changing traffic conditions. Encoding all these requirements into a quantum optimization model can increase the number of variables and constraints.

The literature therefore suggests that hybrid approaches are currently more practical than attempting to solve the complete transportation problem using a quantum processor alone. A classical system can handle data processing and parts of the optimization while the quantum component can be applied to the combinatorial optimization stage.

IV CRITICAL ANALYSIS OF REVIEWED LITERATURE

The characteristics of the selected studies are summarized in Table VII. The reviewed literature shows that quantum computing has been investigated for transportation problems from several different perspectives. However, the studies do not address the complete smart delivery routing problem in a single framework.

The PAQS study mainly addresses the data-processing side of the problem. It investigates real-time Big Data analytics and demonstrates the importance of distributed and streaming processing for continuously generated data [2]. However, its primary focus is Big Data analytics rather than multi-vehicle delivery route optimization.

The IQGA study considers the routing side of transportation while also incorporating dynamic traffic conditions. Its quantum genetic algorithm evaluates alternative routes using traffic and route-related factors [3]. An important limitation is that the quantum operations are simulated using classical computing resources. Therefore, the study demonstrates the potential

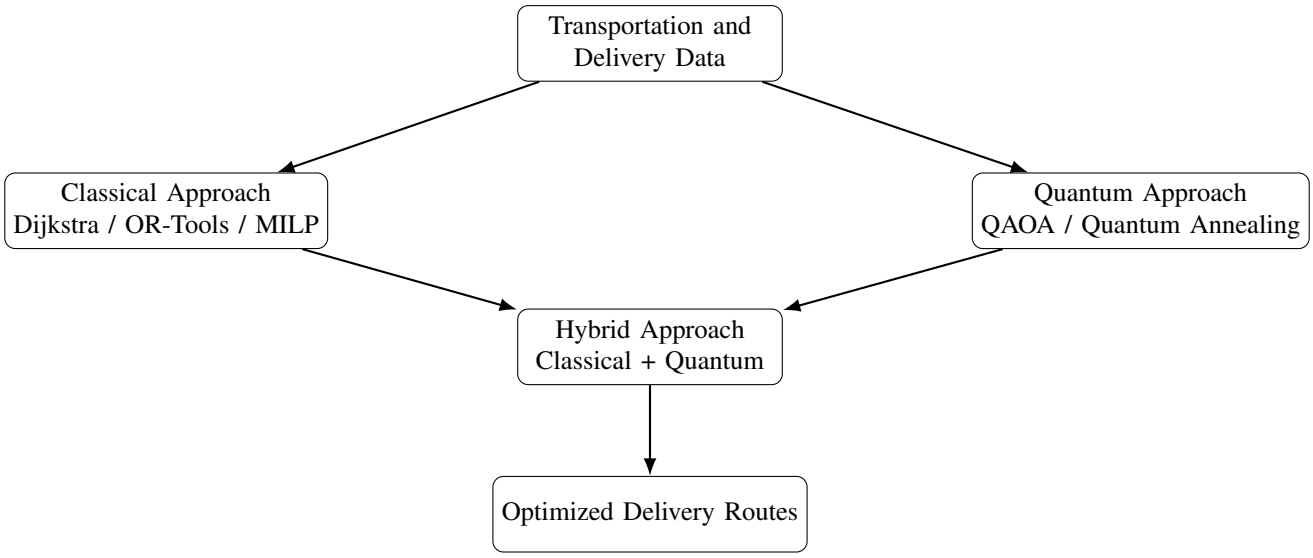


Fig. 5. Comparison of classical, quantum, and hybrid approaches for transportation route optimization.

TABLE VII
CHARACTERISTICS OF THE SELECTED RESEARCH AND SURVEY PAPERS

Ref.	Title / Study	Year	Problem	Algorithm / Method	Technology / Resource
[1]	PAQS: Advanced Optimization for Big Data Streams with Quantum Insights for Real-time Big Data Analytics	2025	Real-time Big Data analytics	Distributed and quantum-inspired optimization	Kafka, Spark
[2]	IQGA: A Route Selection Method Based on Quantum Genetic Algorithm	2018	Dynamic urban routing	Quantum Genetic Algorithm	Classical simulation
[3]	Solving Vehicle Routing Problem Using Quantum Approximate Optimization Algorithm	2023	VRP	QUBO, Ising, QAOA	Qiskit / quantum simulation
[4]	Quantum Computing in Transportation Engineering: A Survey	–	Transportation optimization	QAOA, quantum annealing, QML and other methods	Multiple quantum platforms
[5]	A Systematic Literature Review of Quantum Computing for Routing Problems	2022	TSP, VRP and routing	Quantum, quantum-inspired and hybrid approaches	D-Wave, IBM and others

of quantum-inspired optimization rather than demonstrating quantum hardware advantage.

The QAOA-based VRP study provides a more direct connection between vehicle routing and gate-based quantum computing. It formulates the VRP as a QUBO problem and transforms it into an Ising representation before applying QAOA [4]. This provides an important methodological basis for applying QAOA to delivery routing. However, the experiments are conducted on relatively small routing instances, which limits conclusions about large-scale real-world delivery systems.

The transportation survey provides a broader analysis of quantum computing applications in transportation. It discusses QAOA, quantum annealing, quantum machine learning, quantum random walks, and other approaches. It also identifies hardware noise, limited qubit availability, scalability, and benchmarking as important challenges [5].

The systematic literature review by Osaba et al. provides further evidence of the development of quantum computing for routing problems. The review analyzed 53 studies and identified TSP and VRP as major research areas. It also reported the use of both pure quantum and hybrid quantum-classical approaches [1].

IV-A Critical Comparison

The reviewed studies reveal three major research directions. First, Big Data techniques address the challenge of collecting and processing large amounts of transportation information. Second, traffic-aware routing methods incorporate changing road conditions into route selection. Third, quantum optimiza-

TABLE VIII
KEY CONTRIBUTION AND LIMITATION OF REVIEWED STUDIES

Study	Key Contribution	Main Limitation
PAQS	Real-time Big Data processing	Not focused on complete delivery VRP
IQGA	Traffic-aware dynamic routing	Quantum process simulated classically
QAOA-VRP	VRP to QUBO and QAOA	Small problem instances
Transportation Survey	Broad transportation quantum review	Current hardware and scalability limitations
Osaba et al.	Systematic review of quantum routing	Need for realistic and scalable applications

tion methods attempt to address the computational complexity of routing problems.

However, these directions are not completely integrated in the reviewed studies. The Big Data approach does not directly solve the complete vehicle routing problem, while the quantum routing studies generally operate on smaller and more controlled problem instances. Similarly, traffic-aware routing and quantum optimization are not consistently combined with continuous real-time data processing.

Another important observation is that current quantum approaches should not automatically be considered superior to classical methods. Quantum routing studies need to be evaluated against strong classical baselines using the same problem constraints, datasets, and evaluation criteria. The systematic review emphasizes the importance of realistic applications and appropriate benchmarking [1].

Therefore, the literature supports the investigation of a hybrid framework in which Big Data technologies are responsible for processing transportation information, traffic conditions are used to calculate dynamic route costs, and quantum optimization is applied to the resulting vehicle routing problem.

V CRITICAL ANALYSIS OF IMPORTANT PAPERS

The important contributions identified from the reviewed literature can be summarized as follows. The selected studies demonstrate a gradual development from transportation data processing and traffic-aware routing toward quantum-assisted optimization.

The PAQS study, *Advanced Optimization for Big Data Streams with Quantum Insights for Real-time Big Data Analytics* (2025), investigates real-time Big Data analytics using streaming and distributed processing approaches and provides a foundation for handling continuously generated transportation-related data [2].

The IQGA study, *A Route Selection Method Based on Quantum Genetic Algorithm Toward Urban Traffic Management Under Big Data Environment* (2018), introduces a quantum genetic algorithm for dynamic route selection and considers traffic-related factors such as travel time, distance, congestion, and road conditions [3].

The QAOA-based routing study, *Solving Vehicle Routing Problem Using Quantum Approximate Optimization Algorithm* (2023), formulates the Vehicle Routing Problem as a QUBO model and applies QAOA using a hybrid quantum-classical optimization procedure [4].

The transportation survey, *Quantum Computing in Transportation Engineering: A Survey*, reviews quantum computing applications in transportation, including QAOA, quantum annealing, quantum machine learning, and other quantum approaches, while identifying current limitations and future directions [5].

Finally, *A Systematic Literature Review of Quantum Computing for Routing Problems* (2022) systematically reviews quantum computing research in routing, covering TSP, VRP, quantum annealing, gate-model approaches, and hybrid quantum-classical methods [1].

Taken together, these five contributions trace a clear progression from data-centric and traffic-aware techniques toward direct quantum formulations of vehicle routing, framed throughout by the broader surveys and systematic reviews of the field.

The IQGA study represents an important step toward intelligent route selection under dynamic traffic conditions. Instead of considering only the physical distance between locations, the study incorporates route-related and traffic-related factors into the evaluation of alternative routes. A Quantum Genetic Algorithm is then used to search for suitable routes [3]. However, the quantum process in the study is simulated using classical computing resources, and therefore it does not demonstrate a direct quantum-hardware advantage.

The QAOA-based VRP study takes a different approach by directly formulating the vehicle routing problem for quantum optimization. The routing decisions are represented using binary variables and the objective and constraints are converted into a QUBO formulation. The QUBO is then transformed into an Ising representation and solved using QAOA with a classical optimizer [4]. This work is particularly relevant to the proposed research because it provides the connection between VRP, QUBO, and QAOA.

The PAQS study addresses another important part of the proposed system: the processing of large-scale and continuously generated data. Its use of streaming and distributed processing demonstrates how Big Data technologies can be used to process information before intelligent optimization is performed [2]. However, its primary focus is real-time Big Data analytics rather than complete multi-vehicle delivery routing.

The transportation survey provides a broader analysis of quantum computing applications in transportation. It covers quantum optimization methods such as QAOA and quantum annealing, as well as quantum machine learning and other approaches. The survey also highlights practical limitations including noise, limited qubit availability, scalability, and difficulties in applying quantum computing to realistic transportation problems [5].

The systematic literature review by Osaba *et al.* provides a wider view of quantum computing for routing problems. It analyzed 53 studies and showed that TSP and VRP are major areas of research. The review also identified both pure quantum and hybrid quantum-classical approaches [1].

V-A Evolution of the Research

The reviewed studies show an evolution in the way transportation optimization is approached. Traffic-aware routing research demonstrates the importance of incorporating changing road conditions into route selection. Big Data research provides methods for processing the large volumes of information required for such dynamic systems. Quantum routing research then provides optimization techniques for computationally difficult routing problems, connecting the two through the Big Data → Traffic-Aware Cost → VRP → QUBO → QAOA pipeline identified earlier in this survey.

This progression suggests that quantum computing should not be considered independently from the data-processing and transportation components. Instead, a hybrid architecture can use classical technologies for data collection and processing, traffic-aware cost estimation, and constraint handling, while using quantum optimization for the computationally difficult routing component.

VI CONCLUSION AND FUTURE WORK

This literature survey reviewed recent research on Big Data processing, traffic-aware routing, vehicle routing, and quantum computing for transportation optimization. The selected studies show that these areas have developed independently but are closely connected in the context of smart delivery systems.

The reviewed Big Data research demonstrates the importance of streaming and distributed processing for handling continuously generated transportation information. The traffic-aware routing study further shows that route selection should consider dynamic factors such as travel time, congestion, distance, and road conditions rather than relying only on static shortest paths.

Quantum computing provides another direction for solving the combinatorial complexity of routing problems. The reviewed QAOA study demonstrates how a Vehicle Routing Problem can be converted into a QUBO formulation and subsequently optimized using QAOA with a classical optimization loop. Quantum annealing provides an alternative optimization approach that has also been explored for routing and transportation problems.

The survey and systematic literature review papers indicate that quantum computing has been investigated for several transportation applications, including TSP, VRP, traffic management, scheduling, and machine-learning-based transportation tasks. However, current quantum technologies continue to face limitations related to noise, decoherence, limited qubit availability, connectivity, scalability, and parameter optimization.

An important observation from the reviewed literature is that many studies focus on individual components of the transportation problem. Big Data studies mainly address data processing, traffic-aware studies address dynamic route selection, and quantum studies primarily address the optimization component. The integration of these components into a realistic delivery routing framework remains an important research opportunity.

The literature also indicates that quantum advantage cannot be assumed simply because a quantum algorithm is used. Quantum solutions should be evaluated against strong classical methods using comparable problem instances, constraints, and performance measures. The systematic review emphasizes the importance of realistic routing problems and appropriate benchmarking [1].

Based on these observations, a hybrid approach combining Big Data processing, traffic-aware cost estimation, multi-vehicle routing, and quantum optimization is identified as a promising research direction. The proposed research will investigate the following workflow:

$$\text{Big Data} \rightarrow \text{Traffic Information} \rightarrow \text{VRP} \rightarrow \text{QUBO} \rightarrow \text{QAOA} \rightarrow \text{Optimiz} \tag{3}$$

Future work will focus on implementing this framework using realistic or benchmark delivery-routing data. A classical routing method will first be developed as a baseline, followed by the construction of a QUBO formulation and its solution using QAOA. Traffic information can subsequently be incorporated to study dynamic route optimization. The performance of the quantum and classical approaches will be evaluated using measures such as total distance, travel time, delivery delay, route cost, and computational time.

For larger experiments, traffic simulation and Big Data technologies can be incorporated to investigate how the proposed approach behaves when transportation information changes over time. This can provide a pathway toward evaluating hybrid quantum-classical optimization under more realistic delivery conditions.

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